

Node.js Machine Test – Real-Time Chat System

Role: Node.js Backend Developer

Duration: 2–3 Hours

Objective: To evaluate the candidate's ability to design and develop a small-scale backend system using Node.js, Express, Socket.IO, and a Database (MongoDB or MySQL) — with a focus on JavaScript logic, real-time events, and clean code structure.

Task Overview

You are required to build a simple real-time chat system where users can register, create or join chat rooms, and send/receive messages in real time. The purpose of this task is to assess your practical backend coding ability, understanding of modern JavaScript, Socket.IO, and database interactions.

Functional Requirements

Use Node.js (Express), Socket.IO, and MongoDB or MySQL. Maintain a clean and modular folder structure as shown below:

```
/src
  /config
  /models
  /routes
  /controllers
  /socket
index.js
```

API Endpoints

- (a) POST /users/register – Register a user with name and email, validate, and save to DB.
- (b) POST /rooms/create – Create a new chat room and return room ID.
- (c) GET /rooms – List all chat rooms.
- (d) GET /messages/:roomId – Fetch messages for a room.

Socket.IO Events

join_room – User joins a chat room.

send_message – User sends a message, save it in DB, broadcast via receive_message.

receive_message – Broadcast event for others in the same room.

disconnect – Log when user disconnects.

Bonus Tasks (Optional)

- Add message pagination (limit 10 per page).
- Add search in messages.
- Show online users in a room.
- Add timestamps.

Technical Expectations

Use modern JS (ES6+), async/await, Express Router, middleware, Socket.IO events, proper schema, and a clean folder structure. Include README with setup instructions.

Submission Guidelines

1. Zip the project folder.
2. Include README.md with setup steps.
3. (Optional) Push to GitHub.
4. Use .env for configuration.
5. Ensure project runs via 'npm install' and 'npm start'.

Expected Deliverable

A working backend that:

- Starts with npm start
- Allows user and room creation via APIs
- Supports real-time messaging
- Has clean folder structure and documentation